

YANchor-4B

Effective Long-Horizon Reasoning in $O(N)$ Time with $O(1)$ Memory

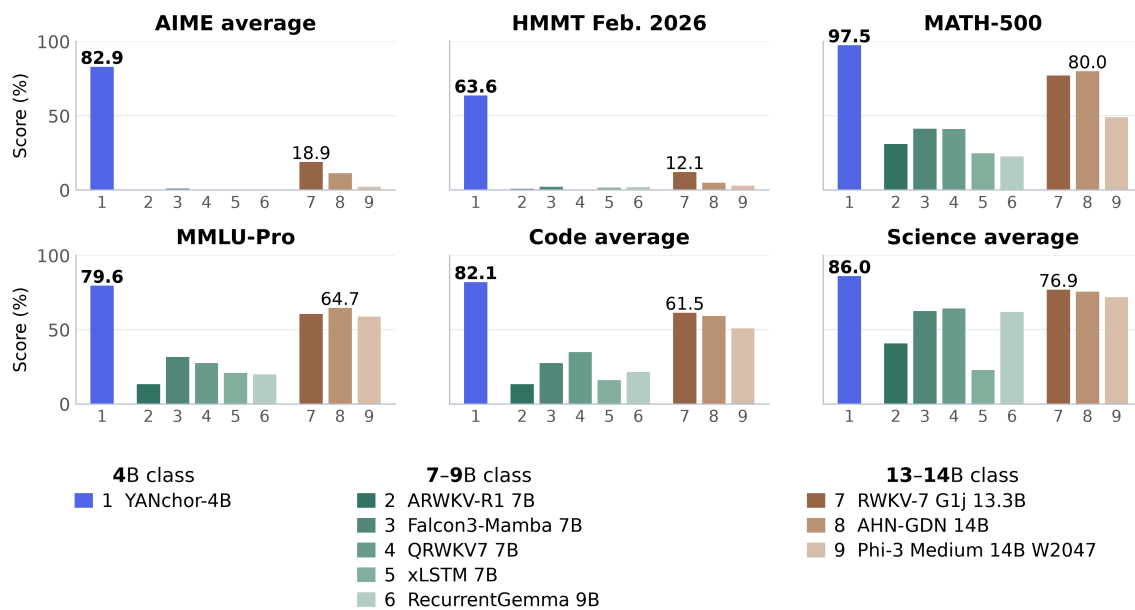
Rocore Matrix · October 2026

[Demo, deployment and evaluation](#)

YANchor-4B is a general-purpose recurrent model that preserves crucial memory as ANchors for retrieval during subsequent reasoning. Its multidimensional memory mechanism enables effective long-horizon reasoning with $O(N)$ generation time and $O(1)$ history-state memory.

For example, on challenging math problems, YANchor achieves 82.93% mean pass@1 on AIME 2024–2026 and 63.64% on HMMT, substantially outperforming linear-time, constant-state counterparts, including 7B–14B models.

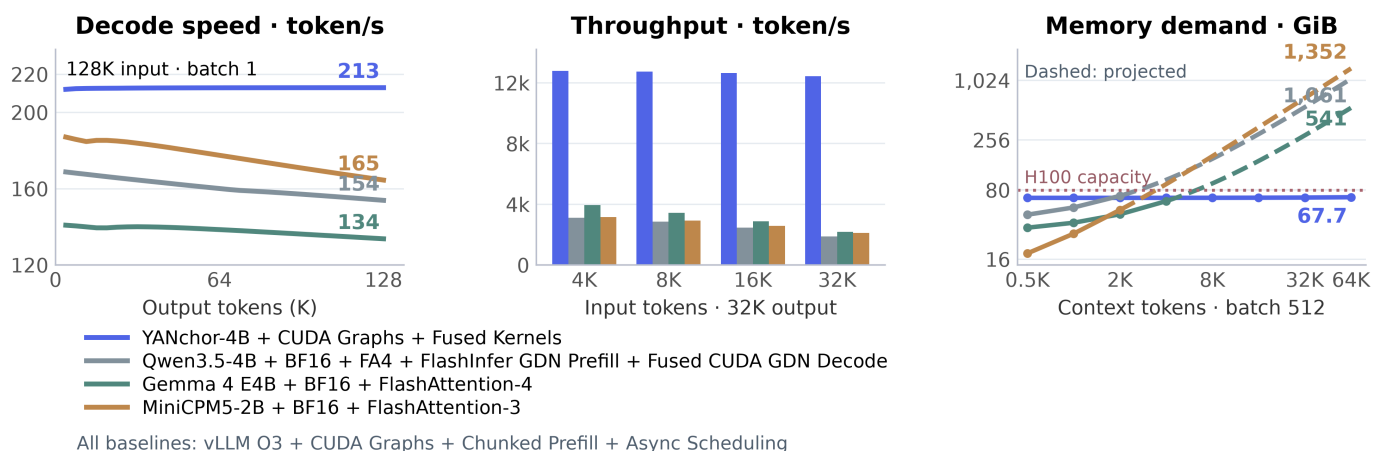
YANchor delivers several-fold higher batched long-generation throughput than Transformer and hybrid baselines on H100. Evaluations across dozens of benchmarks demonstrate its superiority in general-purpose capabilities.



AIME averages 2024–2026; HMMT uses February 2026. Code averages HumanEval, MBPP, and LiveCodeBench v6; science averages GPQA-Diamond, ARC-C, and ARC-E. Benchmarks have equal weight within each aggregate.

[Evaluation settings.](#)

Single-H100 inference: speed, throughput, and memory.



Throughput uses repeated prompts and 32K outputs. Solid memory curves use measured components; dashed curves project KV growth. [Inference settings](#) · [Memory comparison](#).

Evaluation

Evaluation covers 27 text benchmarks and six VL benchmarks. YANchor reports mean pass@1 for correctness-based tasks; Evaluation Settings defines the metrics. Bold marks row-best scores, including ties.

Linear-time, constant-state models

YANchor leads all 16 evaluated linear-time, constant-state baselines on 20 of the 24 common benchmarks. Its 24-benchmark mean is **78.64**, exceeding the strongest baseline, RWKV-7 G1j 13.3B at 56.29, by **22.35 points**. On AIME and HMMT, YANchor scores 82.93 and 63.64, compared with 18.89 and 12.12 for that 13.3B model. The lead spans 7B–14B counterparts and extends to knowledge, code, and instruction following.

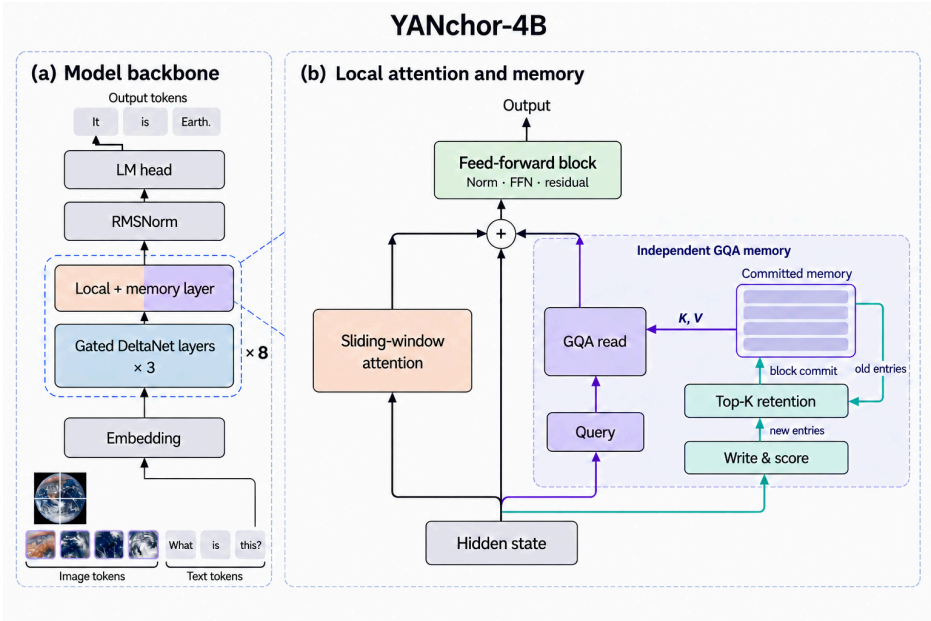
Benchmark	YANchor-4B	ARWKV-R1 7B [2]	Falcon3-Mamba 7B Instruct [4]	QRWKV7 7B Instruct [5]	xLSTM 7B [6]	Recurrent Gemma 9B IT [8]	RWKV-7 G1j 13.3B [7]	AHN-GDN Qwen2.5 14B [3]	Phi-3 Medium 4K Instruct W2047 [9]
Mathematics									
AIME 2024 [12]	86.72	0.00	1.67	0.00	0.00	0.42	17.50	13.75	3.33
AIME 2025 [12]	77.45	0.00	0.83	0.00	0.00	0.42	27.92	10.00	3.33
AIME 2026 [12]	84.64	0.42	0.83	0.00	0.00	0.00	11.25	10.42	0.00
HMMT Feb 2026 [13]	63.64	0.76	2.27	0.00	1.52	1.89	12.12	4.92	3.03
MATH-500 [11]	97.49	31.00	41.20	41.00	24.80	22.60	77.00	80.00	49.00
GSM8K [10]	94.45	44.50	60.60	65.00	41.60	51.80	94.30	93.10	84.60
Knowledge									
MMLU [14]	87.87	36.10	61.50	66.20	23.50	56.00	76.80	77.30	75.90
MMLU-Pro [15]	79.60	13.30	31.70	27.60	21.00	19.90	60.60	64.70	58.80
MMLU-Redux [16]	92.00	36.69	64.26	35.32	38.99	54.82	79.66	83.33	81.45
CMMLU [17]	72.40	29.47	35.28	57.90	19.35	40.42	66.14	75.78	53.60
C-Eval [18]	74.47	28.60	39.00	57.20	19.20	40.70	66.60	76.40	50.90
Science									
GPQA-Diamond [19]	64.61	13.64	24.24	10.61	12.63	25.25	42.42	40.91	29.80
SuperGPQA [20]	61.69	9.90	12.30	16.90	5.90	19.30	35.50	36.30	25.30
ARC-C [23]	95.50	50.30	76.70	86.00	27.10	73.20	91.60	90.90	90.20
ARC-E [23]	98.02	58.10	86.90	96.00	29.00	87.40	96.60	95.10	95.50
Commonsense and reasoning									
WinoGrande [21]	77.90	36.50	59.90	61.80	25.80	56.40	72.50	77.20	80.60
HellaSwag [22]	64.57	12.70	41.00	45.20	9.20	44.90	76.40	70.00	76.10
BBH [24]	85.88	12.00	31.50	37.39	10.90	35.30	48.89	53.57	51.87
BBEH [25]	29.07	2.11	3.51	1.91	3.32	2.41	7.64	4.50	2.61
DROP F1 [31]	77.22	10.38	19.87	21.02	30.71	38.25	54.38	65.32	50.30
MuSR [32]	60.42	10.96	23.34	1.84	10.92	35.04	51.35	56.13	50.55
LogiQA2 [33]	72.23	21.10	40.20	26.10	18.00	40.40	66.70	70.70	61.50
Code									
HumanEval [26]	96.72	16.46	34.15	61.59	36.59	37.20	85.98	75.61	73.78
MBPP [27]	87.81	21.20	39.20	30.20	9.20	19.00	67.40	68.60	59.40
LiveCodeBench v6 [28]	61.63	2.60	9.50	12.70	2.30	8.20	31.00	33.30	19.60
Instruction following									

Benchmark	YANchor-4B	ARWKV-R1 7B [2]	Falcon3- Mamba 7B Instruct [4]	QRWKV7 7B Instruct [5]	xLSTM 7B [6]	Recurrent Gemma 9B IT [8]	RWKV-7 G1j 13.3B [7]	AHN-GDN Qwen2.5 14B [3]	Phi-3 Medium 4K Instruct W2047 [9]
IFEval [29]	88.03	10.35	64.70	58.78	35.49	39.56	77.82	77.26	61.55
IFBench [30]	65.29	8.33	18.67	21.33	14.67	12.67	27.33	33.33	22.00

Model

Model	Text parameters	Weights	Model license
YANchor-4B	4.58B Hugging Face		YANchor Model License 1.0

Architecture



Long-horizon reasoning needs access to crucial information after it leaves the local context. YANchor preserves this information as persistent memory anchors, making distant context available for retrieval during subsequent reasoning.

YANchor retains Qwen3.5-4B's recurrent layers and replaces its global-attention positions with local attention and independent memory. Three complementary paths carry earlier constraints, intermediate results, and exact associations.

Information path	Role
Gated DeltaNet	Maintains a fixed-size recurrent summary of history.
Sliding-window attention	Resolves detailed interactions among recent tokens.
Independent GQA memory	Learns what to write, which records to retain, and how to retrieve distant information.

The memory has independent writing, admission, and retrieval parameters. New records compete with retained entries through blockwise Top-K selection, which preserves the written key–value representations of the retained records. Current queries read these memory anchors through grouped-query attention. Local and memory attention are normalized separately, and their outputs join the residual stream.

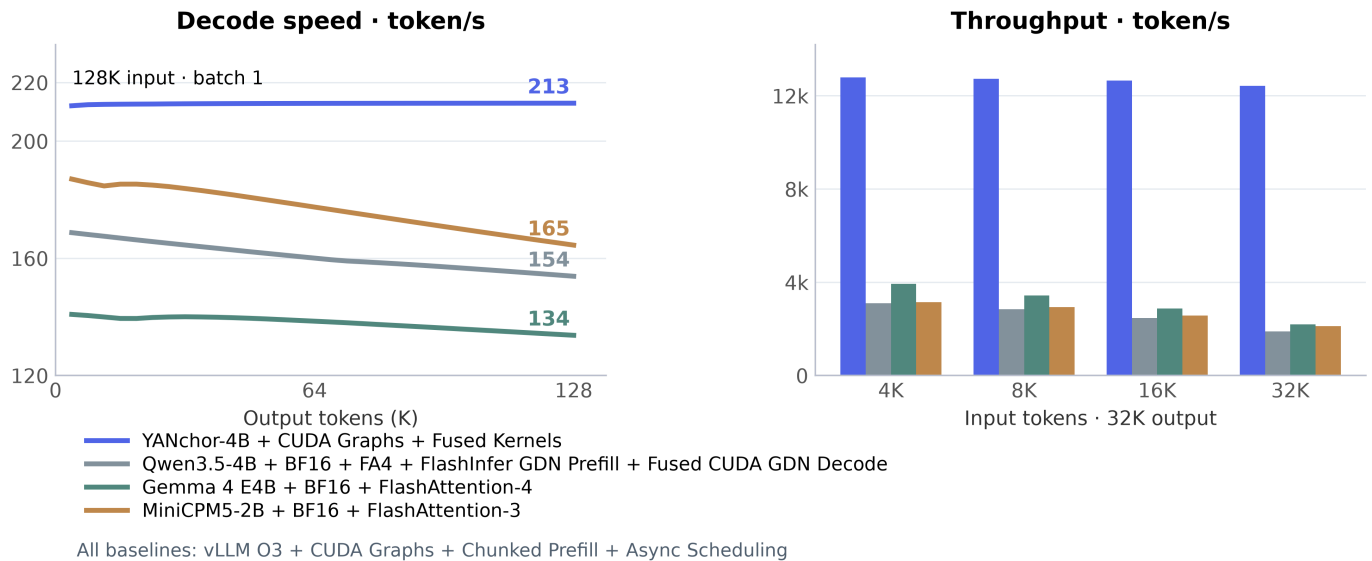
The recurrent state, local window, memory bank, candidate records, and metadata have fixed capacity. YANchor therefore generates N tokens in O(N) time using O(1) history-state memory.

Inference efficiency

Fixed-capacity history supports sustained decoding and larger active batches. All comparisons use one H100 80GB.

Single-sequence speed. YANchor sustains 212.98 cumulative decode tokens/s at 128K input + 128K output; Qwen3.5-4B, Gemma 4 E4B, and MiniCPM5-2B reach 153.91, 133.75, and 164.53 tokens/s, respectively. YANchor's curve remains nearly flat as generation extends.

Batched throughput. YANchor delivers 5.66–6.58 times the throughput of Gemma, MiniCPM, and Qwen at 32K input + 32K output. It reaches 12,431 end-to-end tokens/s, compared with 2,196, 2,117, and 1,889, respectively.



Inference configurations

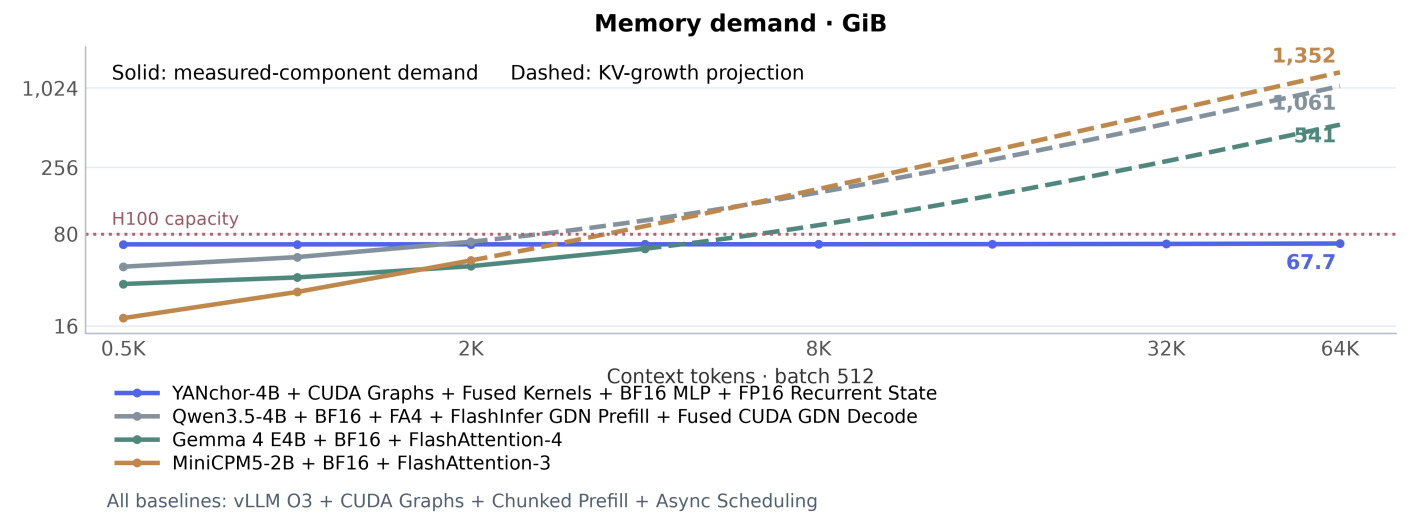
Tests use synthetic token inputs and fixed-length generation with EOS ignored; tokenizer and text-decoding costs are excluded. Batch tests use repeated prompts with independent continuation states. Batch candidates are 32–512 for YANchor, 64/128 for Gemma, and 32/64 for Qwen and MiniCPM; the comparison reports each model's highest measured throughput. Tensor-parallel size is one. Pure decode is timed after first-token delivery; end-to-end throughput covers the complete generation call, including prefill, queueing, scheduling, and cache-preemption recomputation. Loading and warmup are outside both intervals. Here 1K is 1,024 tokens.

YANchor uses CUDA Graphs and fused kernels, with INT8 MLP / FP32 recurrent state at B1 and BF16 MLP / FP16 recurrent state in batch tests. Qwen uses BF16, FlashAttention-4, FlashInfer GDN prefill, and fused CUDA GDN decode. Gemma uses BF16 and full-layer FlashAttention-4; MiniCPM uses BF16 and FlashAttention-3. All three baselines enable vLLM O3, CUDA Graphs, chunked prefill, and asynchronous scheduling.

Gemma and MiniCPM extend runtime and position-table capacity to 256K for the B1 speed measurement, keeping weights and RoPE frequencies unchanged. Deployment references: [Gemma/vLLM](#), [MiniCPM5](#).

GPU memory

YANchor keeps 512 independent long-context sequences resident on one H100 80GB. Its history cache stays fixed at 48.90 GiB as context grows from 512 to 65,536 tokens per sequence. Peak memory demand remains near 67 GiB; the 1 GiB increase comes from input buffers.



Each sequence retains an independent history. Solid curves reconstruct peak demand from measured occupied cache, workspace, and runtime overhead, excluding unused preallocated cache. Dashed curves project full-batch demand from the last successful point. The device has 79.65 GiB of memory.

Cache preemption begins at 4K context for Qwen and MiniCPM, and at 8K for Gemma. Their projected demands at 64K are 1,061, 1,352, and 541 GiB, respectively. YANchor's fixed history storage enables the longer contexts and larger active batches that drive its throughput.

Long-memory capability

YANchor retrieves and uses distant information with 94.14% query accuracy over 16K–130,000-token contexts. The tasks cover exact retrieval, state updates, cross-document combination, and variable binding.

Memory path	Query accuracy	All four answers correct
Enabled	94.14%	80.08%
Disabled	0.39%	0.00%

Disabling memory reduces query accuracy to 0.39%. Paired evidence-change tests further show that YANchor uses distant content to update its answers. The technical report gives the task breakdown and experimental settings.

Transformer and global-attention hybrid references

These models provide cross-architecture references with context-dependent history storage. Qwen3.5-4B combines Gated DeltaNet with global attention and serves as the backbone reference for YANchor's conversion to bounded-state inference.

YANchor surpasses MiniCPM4.1-8B, Gemma4-E4B, Phi-4-mini-reasoning, and Nemotron-3-Nano-4B on all three AIME editions while using constant history-state memory. Compared with Qwen3.5-4B, YANchor scores higher on MMLU, MMLU-Pro, MMLU-Redux, SuperGPQA, HumanEval, MBPP, and IFBench. Across the 24 common benchmarks, YANchor and Qwen score 78.64 and 80.44, respectively. YANchor retains 96.31% of Qwen's three-year AIME mean while using 43–47% fewer output tokens, and delivers 4.12–6.58 times its measured end-to-end throughput in batched long generation.

Benchmark	YANchor-4B	Qwen3.5-4B Backbone [1]	Granite4.2 3B [41]	Phi-4-mini-reasoning [42]	Gemma4 E4B-it [40]	Nemotron-3 Nano-4B [43]	MiniCPM4.1 8B [39]
Mathematics							
AIME 2024 [12]	86.72	89.58	85.00	44.58	41.25	60.83	77.08
AIME 2025 [12]	77.45	80.83	78.75	30.83	39.58	56.67	70.00
AIME 2026 [12]	84.64	87.92	82.92	35.00	38.33	53.75	70.83
HMMT Feb 2026 [13]	63.64	65.91	58.71	26.14	28.03	36.36	50.00

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MATH-500 [11]	97.49	98.00	97.80	91.40	87.40	94.80	96.40
GSM8K [10]	94.45	94.50	93.50	92.40	90.30	91.80	93.30
Knowledge							
MMLU [14]	87.87	85.60	77.30	78.10	75.20	77.30	82.20
MMLU-Pro [15]	79.60	75.90	65.80	53.20	68.60	56.40	65.90
MMLU-Redux [16]	92.00	89.31	80.92	79.77	85.95	80.82	86.79
CMMLU [17]	72.40	82.26	58.29	44.34	61.84	53.94	80.12
C-Eval [18]	74.47	85.10	61.80	42.70	64.90	55.70	78.80
Science							
GPQA-Diamond [19]	64.61	77.78	58.59	47.98	40.91	43.43	50.51
SuperGPQA [20]	61.69	51.40	39.60	34.70	42.50	37.50	41.90
ARC-C [23]	95.50	96.90	92.80	91.40	92.10	91.60	92.70
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WinoGrande [21]	77.90	87.00	74.60	75.90	76.40	65.30	72.70
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BBH [24]	85.88	85.81	86.22	74.92	85.10	80.00	88.10
BBEH [25]	29.07	34.31	25.10	12.26	16.60	9.22	12.79
DROP F1 [31]	77.22	75.41	83.35	54.05	72.58	71.95	78.60
MuSR [32]	60.42	64.78	65.07	52.45	62.18	58.24	64.49
LogiQA2 [33]	72.23	81.30	68.70	57.60	73.50	67.30	76.80
Code							
HumanEval [26]	96.72	85.98	96.95	81.71	93.90	90.85	98.17
MBPP [27]	87.81	70.80	92.60	64.00	79.20	72.20	92.20
LiveCodeBench v6 [28]	61.63	76.20	76.30	35.80	69.50	67.10	70.80
Instruction following							
IFEval [29]	88.03	90.02	92.42	46.03	84.29	89.28	73.75
IFBench [30]	65.29	60.33	74.33	14.67	34.67	57.67	23.67

VL comparison

YANchor supports bilingual VL question answering, object-presence judgments, visual perception, chart understanding, and scientific-diagram reasoning.

Benchmark	YANchor-4B	Qwen3.5-4B [1]
Vision-language		
MMBench EN [34]	88.28	90.08
MMBench CN [34]	89.00	91.43
POPE [35]	85.43	88.76
MME [36]	82.98	86.82
ChartQA [37]	83.92	90.28
AI2D [38]	80.05	86.98

VL scores are answer-level accuracy in percent.

Evaluation Settings

YANchor mean pass@1. For text evaluation, YANchor sets the number of independent responses per item, K , by evaluation-set size: $K=64$ for fewer than 100 questions, $K=16$ for 100–10,000 questions, and $K=1$ for more than 10,000 questions. More repetitions on smaller sets reduce variation from stochastic generation. Binary-correctness scores average across responses and then across problems, estimating single-response success. Other metrics retain their benchmark-specific definitions.

Scoring. Mathematics uses final-answer correctness. YANchor's code scores count extracted programs that pass the task tests, including code recovered from the end of reasoning. Knowledge and instruction-following tasks use their benchmark-specific aggregation and checkers.

Generation settings

YANchor's text evaluation uses temperature 0.6, top-p 0.95, top-k 20, and a 131,072-token response budget. Completions beyond this budget receive zero outcome credit. VL evaluation uses one response per input with thinking enabled, temperature 1.0, top-p 0.95, top-k 20, presence penalty 1.5, and a 32,768-token response cap.

License

YANchor-4B weights use [YANchor Model License 1.0](#), a custom license incorporating Apache License 2.0 terms. The sole additional restriction is physically hardwiring covered weights into a chip's non-reprogrammable structure, including commissioning such fabrication, without written authorization. Research, training, fine-tuning, distillation, quantization, deployment, redistribution and commercial use are permitted without fees or scale thresholds. Ordinary device deployment, model-data storage and hardware design remain permitted; read-only storage alone does not trigger the restriction. Code uses [Apache-2.0](#). Upstream and third-party materials retain their original licenses; see [NOTICE](#).

Citation

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  author = {Ji, Huishan and Xu, Hua and Zhang, Weiming and Ye, Qirui},  
  year  = {2026},  
  url   = {https://github.com/RocoreMatrix/YANchor}  
}
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Citation metadata is available in [CITATION.cff](#).

Contacts

Huishan Ji · jihs@stonehill-tech.com

Hua Xu · xuh@stonehill-tech.com

Weiming Zhang · zhangwm@stonehill-tech.com

Qirui Ye · qiruiy@andrew.cmu.edu

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